

Selecting Thermal Imagers for Electronics Professionals

Infrared cameras are used for capturing and recording thermal distribution and variation in real time, allowing engineers to observe and accurately measure heat patterns, dissipation, leakage and other temperature factors in products. How and why? Read on to find out

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Thermography, or thermal imaging, is a technique of using the heat released by an object to produce an image of it or to locate it. Basically, this method is used to improve visibility of objects, especially in dark. On a broader note, it is used for studying the heat dissipation patterns, locate errors and optimise design of a system.

With miniaturisation of components being one of the key focusses in electronics industry, the heat transfer requirements have also become more stringent. Each component creates some amount of heat during its work cycle. If this heat is not dissipated properly, it can destroy the component and/or the elements adjacent to it. This would lead to reduction in performance or even destruction of the entire equipment in the long run. So it is extremely important to examine the temperature conditions, and analyse the warming and cooling behaviour of individual components and the system as a whole, over definite time intervals.

This technology is far more advanced and useful for PCB industry. "Many people use spot meter to check if something is getting heated up. This is dependent upon the person doing the test and his capability without any data storage for later review," says T.P. Singh, director-sales (HH), FLIR Systems India Pvt Ltd. All these problems are sorted out with one click of thermal camera.

Benefits. Infrared cameras are being widely used in production and diagnostics in electronics industry. Ther-

mography or thermal infrared imaging serves as an efficient tool for increasing the productivity and life expectancy of electronics devices. It gives an accurate picture of real-time, contact-less, temperature distribution and development of small, irregular-shaped objects. Effective thermal imaging analysis helps optimise the dimensioning and

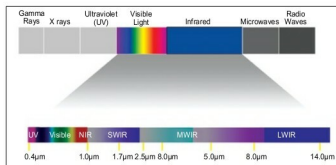
Their use is limited only by an instructor's imagination," explains Singh.

Latest trends in the market

A general trend found in thermal imagers market is the decrease in price, making it a viable, cost-effective tool. Thermographers are being manufactured in low resolution, making the price low enough to be adopted by technicians and DIY enthusiasts.

"Thermography, till a few years ago, was considered a technology to be used by experts.

However, with manufacturers



The electromagnetic spectrum

positioning of circuit components and their assemblies. This method helps to identify defects in circuit board assembly, such as short circuit or stress, that cannot be identified easily using the conventional techniques of troubleshooting and testing.

"Infrared cameras make accurate non-contact temperature measurements on a wide range of objects. For every situation characterised by a heat transfer mechanism, may it be conductive, convective or radiative, infrared cameras are used for quantitative and/or qualitative evaluation on research projects, from swirling flow dynamics to the thermo-mechanical behaviour of composite structures. Because of their dramatic visual presentation of heat energy and temperature, IR cameras strongly reinforce the learning process.

coming into the market with innovative features to make the usage of the camera easy, it is now being used effectively by people with no training in thermography," explains Milind Thakkar, marketing manager, Fluke.

The devices are now smaller in size, less expensive and consume less power, thereby increasing the adoption of this technology in a wide range of commercial applications.

Image quality is also one of the focusses. Traditionally, the quality of thermograms was dependent on several environmental factors. However, in newer products, environmental influences like rain, fog, sunlight, etc do not have substantial effects on the quality of thermograms.

"Recently a visual light camera module was introduced in our thermal